Chemical Institute of Canada

ASDAN China 阿思丹学院

JUNIOR CANADIAN CHEMISTRY OLYMPIAD

JCCO 2020

Instructions

The Periodic Table/Data Sheet may be used for all parts of the contest, and non-programmable calculators are allowed.

考试期间允许使用化学元素周期表以及非编程计算器。

Write your answers on the Multiple Choice Answer Sheet provided. ***Use a 2B pencil.***

请将答案用 **2B 铅笔**正确填涂在答题卡上。

While students are expected to attempt all questions for a complete examination in 1.5 hours.

考试时长为 90 分钟。

1) Low-dose aspirin ($C_9H_8O_4$)is known to prevent heart attacks. Low-dose aspirin often comes in 81 mg tablets. If a person consumes one 81mg tablet, how many moles of aspirin would they be consuming?

低剂量的阿司匹林($C_9H_8O_4$)通常为 81 mg 的片剂，可以预防心脏病。如果一个人吃掉一片 81 mg 的片剂，那他会消耗多少摩尔的阿司匹林？

a) 2.22×10^3 b) 4.50×10^{-4} c) 4.50×10^{-1} d) 1.5×10^{-3} e) 2.22

2) Which of the following chemical equations is properly balanced?

以下化学方程式配平正确的是哪个?

a) $4C_3H_5N_3O_9 \rightarrow 12CO_2 + 10H_2O + 6N_2 + O_2$
b) $C_6H_8 + 6O_2 \rightarrow 5CO_2 + 4H_2O$
c) $CH_3CH_2OH + 3O_2 \rightarrow CO_2 + 3H_2O$
d) $CH_3OH + O_2 \rightarrow CO_2 + 2H_2O$
e) $2C_7H_5N_3O_6 \rightarrow 2N_2 + 5H_2O + 7CO + 7C$

3) Calculate the mass of sodium chloride (NaCl) required to prepare a 3.00 M solution in water using a 200 mL volumetric flask?

在 200mL 容量瓶中制备 3.00 M 的 NaCl 水溶液，计算需要氯化钠（NaCl）的质量是多少？

a) 13.80 g b) 34.80 g c) 35.06 g d) 21.27 g e) 3.51 g

4) The density of benzene (C_6H_6) at 20 ºC is 0.8765 g/cm^3. What is the volume of 1 mole of benzene at 20 ºC?
20 ºC 时苯（C_6H_6）的密度为 0.8765 g/cm^3。在同样温度条件下，1 摩尔苯的体积是多少？

a) 89.1 L b) 68.5 mL c) 36.4 mL d) 68.5 L e) 89.1 mL

5) Classify each observation as a physical or a chemical property and tally them.

将下列每个观察结果按照物理或化学性质进行分类并记录。

Observation 1: Bubbles form on a piece of metal when it is dropped into acid.
观察结果 1：金属遇酸表面会生成气泡。
Observation 2: The color of a crystalline substance is blue.
观察结果 2：某结晶物质的颜色为蓝色。
Observation 3: A shiny metal melts at 820°C.
观察结果 3：有光泽的金属熔点为 820℃。
Observation 4: The density of a solution is 1.34 g/cm^3.
观察结果 4：某溶液密度为 1.34 g/cm^3 。

a) 3 chemical properties and 1 physical property
   3 个属于化学性质，1 个属于物理性质
b) 2 chemical properties and 2 physical properties
   2 个属于化学性质，2 个属于物理性质
c) 1 chemical property and 3 physical properties
   1 个属于化学性质，3 个属于物理性质
d) 4 physical properties
   4 个都属于物理性质
e) 4 chemical properties
   4 个都属于化学性质

6) Several general types of chemical reactions can occur based on what happens when going from reactants to products. What is the type of the following reaction?

$$CH_4 + 2O_2 \rightarrow 2H_2O + CO_2$$

根据从反应物到生成物所发生的变化，可将化学反应分为几类。

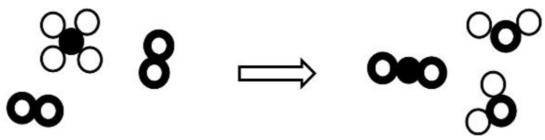
$CH_4 + 2O_2 \rightarrow 2H_2O + CO_2$ 属于下列哪一类型的化学反应？

a) Synthesis
   化合反应
b) Neutralization
   中和反应
c) Decomposition
   分解反应
d) Single replacement
   置换反应
e) Combustion
   氧化反应

7) What is the correctly balanced form of the chemical reaction depicted in the figure below?

关于下图所示化学反应，方程式配平正确的是哪一个？

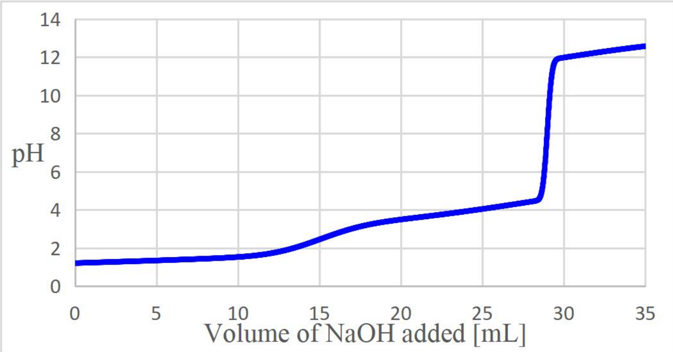
Element A（元素 A）：●
Element B（元素 B）：○
Element C（元素 C）：◯



a) $AB_4 + 4C \rightarrow AB_2 + C_4B_2$
b) $AB_4 + 2C_2 \rightarrow AC_2 + 2CB_2$
c) $4BA + 2C \rightarrow AB_2 + 2CB_2$
d) $AB_4 + 4C \rightarrow AB_2 + 2C_2B$
e) $4BA + 2C_2 \rightarrow 2AB_2 + 2C_2B$

8) A chemist titrated aqueous oxalic acid with sodium hydroxide and recorded the pH on the graph below. Estimate the pH at the last equivalence point observed for this titration. Use the titration curve below.

一名化学家用氢氧化钠溶液滴定草酸溶液，并记录了该过程的 pH 变化，如下图所示。利用下面的滴定曲线图，估算在此滴定过程中到达第一个化学计量点时的 pH 值。



加入 NaOH 的体积 [mL]

a) 1.0
b) 4.0
c) 8.0
d) 13.0
e) 29.0

9) Chemical oxygen generators can provide oxygen for breathing in case of emergency in aircrafts and submarines. The oxygen source is usually an inorganic superoxide, chlorate or perchlorate. If 138.5 grams of potassium perchlorate is decomposed inside an oxygen generator, ($KClO_4 \longrightarrow KCl + 2O_2$). How many grams of dioxygen are produced? (assume a yield of 100% for the above reaction)

化学氧气生成器可以在紧急情况下为飞机和潜艇提供氧气。氧气来源通常为无机过氧化物、氯酸盐或者高氯酸盐。如果氧气生成器中分解了 138.5 克高氯酸钾（$KClO_4 \longrightarrow KCl + 2O_2$），那么会生成多少克的氧气？（假设上述化学反应的产率为 100%）

a) 8 b) 16 c) 32 d) 64 e) 128

10) Assuming ideal gas behavior for the reaction in question 9, at 25°C and 1 bar, what would be the volume of dioxygen (O_2) produced?

假设第 9 题中的化学反应为理想气体状态，即 25℃，1 bar，那么生成的氧气（O_2）体积是多少？

a) 24.8 L b) 49.6 L c) 99.2 L d) 49.6 mL e) 24.8 mL

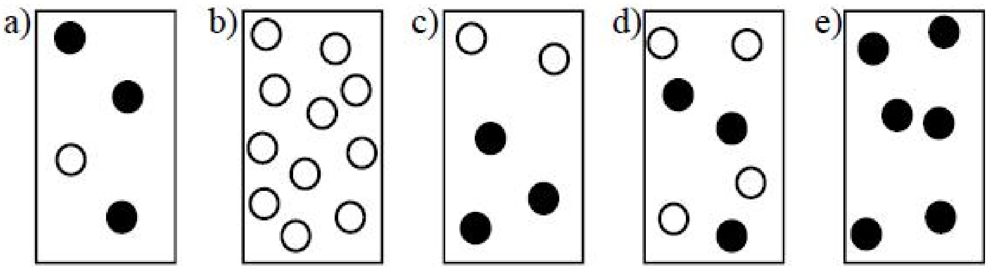
11) The *molality (m)* of a solution is defined as the number of moles of solute per kilogram of solvent. The dissolution of chemical saltpetre (potassium nitrate, KNO_3) in water is a strongly endothermic reaction. This chemical process has been used in Middle Ages to chill water, wine and other beverages indirectly. If a basin containing 60 kg of water and 6 kg of saltpetre were used to cool drinks, what would be the *molality* of the KNO_3 solution?

溶液的*质量摩尔浓度*（*m*）即每千克溶剂中溶质的物质的量。化学硝石（硝酸钾, KNO_3）溶于水是吸热反应。在中世纪，该化学反应曾被用于冷却水、酒和其他饮品。如果使用装有 60 kg 水和 6 kg 硝酸钾的水盆来冷却饮料，那么 KNO_3 溶液的*质量摩尔浓度*是多少？

a) 0.989 *m*　　b) 1.978 *m*　　c) 0.495 *m*　　d) 1.177 *m*　　e) 5.982 *m*

12) For a reaction A + B ⟶ Products, the rate law is: $v = k[A]^2[B]$. For gas containers with the same volume, which mixture would have the highest rate of reaction according to this rate law?
对于一个化学反应 A + B→生成物，速率定律为：$v = k[A]^2[B]$。根据此定律，对于体积相同的气体容器，下列哪种混合物的反应速率最快？

Legend: compound A ○
图示： 化合物 A
compound B ●
化合物 B



13) Two moles of liquid water initially at 274 K are mixing with two moles of liquid water initially at 362 K in a perfectly insulated container. Assume that the molar heat capacity of liquid water is constant and independent of temperature.Calculate the final equilibrium temperature of the water after mixing.

在完全绝热的容器中，将 2 mol 初始状态为 274K 的液态水与 2 mol 初始状态为 362K 的液态水进行混合。假设液态水的摩尔热容恒定且与温度无关。计算混合后水的最终平衡温度。

a) 298K　　b) 308K　　c) 318 K　　d) 328 K　　e) 358 K

14) In the compound X_2O_3, the electron configuration of the element X is $1s^22s^22p^63s^23p^63d^3$. Which of the following element correspond to X ?

在化合物 X_2O_3 中，元素 X 的核外电子排布式为 $1s^22s^22p^63s^23p^63d^3$。以下哪个元素是元素 X?

a) Al b) V c) Cr d) Sc e) Fe

15) Which of the following correctly lists the order of **increasing** electrical conductivity of an aqueous solution by dissolving 1.0 mole of following substances respectively into water to make 1.0 liter of solution?

将 1.0 mol 选项中的物质分别溶解于水，以制备 1.0 L 溶液，水溶液电导率**从小到大**排列正确的是下列哪一项?

a) $CH_3OH < CH_3COOH < AgCl < CaCl_2 < HCl$
b) $CH_3COOH < CH_3OH < CaCl_2 < HCl < AgCl$
c) $CH_3OH < AgCl < CH_3COOH < HCl < CaCl_2$
d) $AgCl < CH_3OH < CH_3COOH < CaCl_2 < HCl$
e) $CH_3OH < CH_3COOH < HCl < AgCl < CaCl_2$

16) Which of the following accurately represents a trend in electronegativity?
以下哪项可以准确表示元素的电负性变化趋势?
a) N>O>F
b) Ne>F>O
c) F>S>O
d) C>N>O
e) Cl>Br >I

17) Which of the following compounds is containing covalent and ionic bonds?

下列哪个化合物既含有共价键又含有离子键?

a) HBr b) KCl c) H_2O d) KNO_3 e) SiO_2

18) One mole of uranium-238 is slowly decaying. If the kinetics of this process is first order, after how many half-lives there is only 0.125 moles of uranium-238 left?

1 mol 的铀-238 正在缓慢衰变。如果此过程遵循一级动力学反应，那么最后只剩下 0.125 mol 的铀-238，中间经历了多少个半衰期?

a) 3 b) 12 c) 8 d) 4 e) 9

19) Archimedes principle states: "When an object is completely submerged, the volume of the displaced water equals the volume of the object." A 200 mL graduated cylinder was filled with exactly 100 mL of water before a 27.0 gram copper block is added to the cylinder. What volume should the cylinder read if the block is completely submerged by water? The density of copper is 8.96 g/cm^3.

阿基米德原理：“如果一个物体完全浸没在水中，它排开水的体积等于物体本身的体积。”现在将 100 mL 的水注入 200 mL 的量筒中，然后在量筒内加入 27.0 g 的铜块。如果铜块完全被水浸没，那么此时从量筒上读取的体积是多少？
已知铜的密度为 8.96 g/cm^3。

a) 103 mL　b) 130 mL　c) 127 mL　d) 142 mL　e) 124 mL

20) A sample of an unknown compound with formula of C_xH_y was combusted with excess amount of oxygen, the reaction produced 132.001g of carbon dioxide and 72.064g of water. Based on the information given, this unknown compound is

一化合物的分子式为 C_xH_y，它在足量的氧气中燃烧，生成 132.001g 二氧化碳和 72.064g 水。根据以上信息可推断，该化合物为

a) CH_4　b) C_2H_6　c) C_3H_8　d) C_4H_{10}　e) C_8H_{18}

21) Which of the following molecules has a tetrahedral molecular geometry?

下列哪个分子具有四面体构型？

a) SF_4　b) CH_4　c) XeF_4　d) BF_3　e) H_2CO

22) How many structural isomers does a compound with molecular formula of C_4H_{10} have?

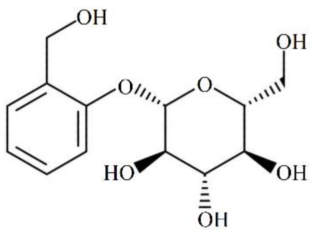
分子式为 C_4H_{10} 的化合物有多少个同分异构体？
a) 6　b) 2　c) 4　d) 3　e) 5

23) A mixture of chromium and zinc weighing 1.000 g was reacted with an excess of hydrochloric acid. After all the metals in the mixture reacted, 618 mL of wet hydrogen gas was collected over water at 25°C and 103160 Pascal. Determine the mass percent of Zn in the metal sample. ($CrCl_3$ and $ZnCl_2$ were produced and the vapour pressure of water at 25°C is 3160Pa)
质量为 1.000 g 的铬锌混合物与过量盐酸进行反应。混合物中的所有金属完全反应后，在 25°C、103160 Pa 的水蒸气上方收集到 618 mL 湿氢气。计算金属样品中 Zn 的质量百分比。（已知产物中还有 $CrCl_3$ 和 $ZnCl_2$，25°C 时水的蒸汽压为 3160 Pa）

a) 22.9%　b) 77.1%　c) 67.4%　d) 28.9%　e) 71.1%

24) The molecule below is sometimes used as a substitute to aspirin. It is found in leaves and bark from the willow tree and has been used for its health effects for at least 2,000 years. Which of the following molecular formula is the correct one for the molecule below ?

下图中的分子有时用作阿司匹林的替代物，广泛存在于柳树的叶子和树皮中，应用于医学领域已有 2000 年的历史。以下哪项是该物质的分子式？



Salicin 水杨苷

a) $C_{13}H_{18}O_7$

b) $C_{12}H_{20}O_7$

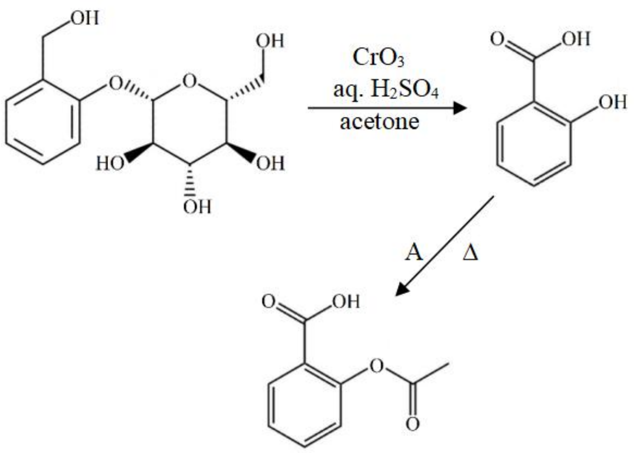
c) $C_{12}H_{16}O_7$

d)$C_{13}H_{13}O_7$

e) $C_{12}H_9O_7$

25) The synthesis of aspirin (acetylsalicylic acid) using salicin as starting material is shown in the reaction scheme below:

利用水杨苷来合成阿司匹林（乙酰水杨酸）的反应过程如下所示：



Which of the following compounds could be used as reagent A?

下列哪种化合物可用作试剂 A？

a) CH_3COOCH_3 b) $CH_3OCH_2CH_3$ c) $CH_3COOCH_2CH_3$ d) $(CH_3CO)_2O$

e) CO_2

End of examination questions.

JUNIOR CANADIAN CHEMISTRY OLYMPIAD

Periodic table and data sheet

1 H 1.008 Hydrogen																	2 He 4.003 Helium
3 Li 6.941 Lithium	4 Be 9.012 Beryllium											5 B 10.81 Boron	6 C 12.01 Carbon	7 N 14.01 Nitrogen	8 O 16.00 Oxygen	9 F 19.00 Fluorine	10 Ne 20.18 Neon
11 Na 22.99 Sodium	12 Mg 24.31 Magnesium											13 Al 26.98 Aluminium	14 Si 28.09 Silicon	15 P 30.97 Phosphorus	16 S 32.07 Sulfur	17 Cl 35.45 Chlorine	18 Ar 39.95 Argon
19 K 39.10 Potassium	20 Ca 40.08 Calcium	21 Sc 44.96 Scandium	22 Ti 47.87 Titanium	23 V 50.94 Vanadium	24 Cr 52.00 Chromium	25 Mn 54.94 Manganese	26 Fe 55.85 Iron	27 Co 58.93 Cobalt	28 Ni 58.69 Nickel	29 Cu 63.55 Copper	30 Zn 65.38 Zinc	31 Ga 69.72 Gallium	32 Ge 72.61 Germanium	33 As 74.92 Arsenic	34 Se 78.96 Selenium	35 Br 79.90 Bromine	36 Kr 83.80 Krypton
37 Rb 85.47 Rubidium	38 Sr 87.62 Strontium	39 Y 88.89 yttrium	40 Zr 91.22 Zirconium	41 Nb 92.91 Niobium	42 Mo 95.96 Molybdenum	43 Tc (98) Technetium	44 Ru 101.1 Ruthenium	45 Rh 102.9 Rhodium	46 Pd 106.4 Palladium	47 Ag 107.9 Silver	48 Cd 112.4 Cadmium	49 In 114.8 Indium	50 Sn 118.7 Tin	51 Sb 121.8 Antimony	52 Te 127.6 Tellurium	53 I 126.9 Iodine	54 Xe 131.3 Xenon
55 Cs 132.9 Cesium	56 Ba 137.3 Barium	57 La 138.9 Lanthanum	72 Hf 178.5 Hafnium	73 Ta 180.9 Tantalum	74 W 183.9 Tungsten	75 Re 186.2 Rhenium	76 Os 190.2 Osmium	77 Ir 192.2 Iridium	78 Pt 195.1 Platinum	79 Au 197.0 Gold	80 Hg 200.6 Mercury	81 Tl 204.4 Thallium	82 Pb 207.2 Lead	83 Bi 209.0 Bismuth	84 Po (209) Polonium	85 At (210) Astatine	86 Rn (222) Radon
87 Fr (223) Francium	88 Ra (226) Radium	89 Ac (227) Actinium	104 Rf (261) Rutherfordium	105 Db (262) Dubnium	106 Sg (266) Seaborgium	107 Bh (264) Bohrium	108 Hs (277) Hassium	109 Mt (268) Meitnerium	110 Ds (269) Darmstadtium	111 Rg (272) Roentgenium	112 Cn (285) Copernicium	113 Nh (284) Nihonium	114 Fl (289) Flerovium	115 Mc (288) Moscovium	116 Lv (292) Livermorium	117 Ts (294) Tennessine	118 Og (294) Oganesson

58 Ce 140.1 Cerium	59 Pr 140.9 Praseodymium	60 Nd 144.2 Neodymium	61 Pm (145) Promethium	62 Sm 150.4 Samarium	63 Eu 152.0 Europium	64 Gd 157.3 Gadolinium	65 Tb 158.9 Terbium	66 Dy 162.5 Dysprosium	67 Ho 164.9 Holmium	68 Er 167.3 Erbium	69 Tm 168.9 Thulium	70 Yb 173.0 Ytterbium	71 Lu 175.0 Lutetium
90 Th 232.0 Thorium	91 Pa 231.0 Protactinium	92 U 238.0 Uranium	93 Np 237.0 Neptunium	94 Pu (244) Plutonium	95 Am (243) Americium	96 Cm (247) Curium	97 Bk (247) Berkelium	98 Cf (251) Californium	99 Es (252) Einsteinium	100 Fm (257) Fermium	101 Md (258) Mendelevium	102 No (259) Nobelium	103 Lr (262) Lawrencium

原子质量单位	Atomic mass unit	1.66054×10^{-27} kg
阿伏伽德罗常量	Avogadro's number	6.022×10^{23}
电子电荷量	Charge of an electron	1.60218×10^{-19} C
法拉第常数	Faraday's constant	96485 C mol-1
理想气体常数	Gas constant	8.3145 J K^{-1} mol^{-1}
电子质量	Mass of an electron	9.1094×10^{-31} kg
光速	Speed of light	2.99792458×10^{8} ms^{-1}
普朗克常数	Plank Constant	6.62637×10^{-34} Js

1 atm = 101325 Pa

1 bar = 100000 Pa

0 ^{o}C = 273.15 K

Draft Paper

Draft Paper